

Customer Shopping Behaviour Analysis Report

1. Project Overview

This project analyzes customer shopping behaviour using transactional data of approximately 3,900 records across various product categories. The main objective is to understand customer purchasing patterns, product preferences, and key factors influencing buying decisions.

The analysis helps identify important trends related to customer segments, product performance, and subscription behaviour, which can support better business strategies.

2. Dataset Summary

- Total Records: 3,900
- Total Columns: 18

Key Features:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Colour
- **Shopping Behaviour:** Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type

Data Quality:

- A total of **37 missing values** were identified in the *Review Rating* column, which were handled during the data preprocessing stage.

3. Exploratory Data Analysis using Python

The data preparation and initial analysis were carried out using Python to ensure the dataset is clean and ready for further processing.

- **Data Loading:** The dataset was imported into the Python environment using the pandas library for analysis.
- **Initial Exploration:** Functions like `df.info()` were used to understand the structure of the dataset, while `df.describe()` helped in analyzing summary statistics of numerical columns.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN

Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
NaN	2	6	2	2	NaN	6	7
NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
NaN	2847	675	2223	2223	NaN	677	584
3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** The dataset was checked for missing values. Missing entries in the Review Rating column were handled by replacing them with the median value of their respective product categories.
- **Column Standardization:** Column names were converted into snake_case format to maintain consistency and improve readability.
- **Feature Engineering:**
 - Created an age_group column by grouping customers into different age categories.
 - Created a purchase_frequency_days column based on customer purchase patterns.
- **Data Consistency Check:** Verified whether columns like discount_applied and promo_code_used contained overlapping information. The promo_code_used column was removed as it was redundant.
- **Database Integration:** The cleaned dataset was then connected to PostgreSQL and stored for further SQL-based analysis.

4. Data Analysis using SQL

We performed structured analysis using PostgreSQL to answer key business questions and extract meaningful insights from the dataset.

1. Revenue by Gender

Compared total revenue generated by male and female customers.

	gender text 🔒	revenue numeric 🔒
1	Female	75191
2	Male	157890

The analysis shows that male customers contributed more to total revenue compared to female customers.

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint 🔒	purchase_amount bigint 🔒
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
Total rows: 839		Query complete 00:00:00.363

This indicates that discounts can encourage higher spending among certain customer groups.

3. Top 5 Products by Rating

Analyzed products with the highest average customer ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Top-rated products reflect strong customer satisfaction and product performance.

4. Shipping Type Comparison

Compared average purchase amounts between standard and express shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Customers using express shipping showed slightly higher spending behaviour.

5. Subscribers vs Non-Subscribers

Compared total customers, average spending, and total revenue based on subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Non-subscribers contributed more to overall revenue, while subscribers showed consistent purchasing patterns.

6. Discount-Dependent Products

Identified products that are most frequently purchased with discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

This highlights products where customers are more price-sensitive.

7. Customer Segmentation

Segmented customers into New, Returning, and Loyal categories based on their purchase behaviour.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

The majority of customers fall under the Loyal segment, indicating strong retention.

8. Top Products per Category

Listed the most purchased products within each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

This helps in identifying high-demand products across categories.

9. Repeat Buyers & Subscriptions

Analyzed whether customers with higher purchase frequency are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Frequent buyers show a higher tendency towards subscription services.

10. Revenue by Age Group

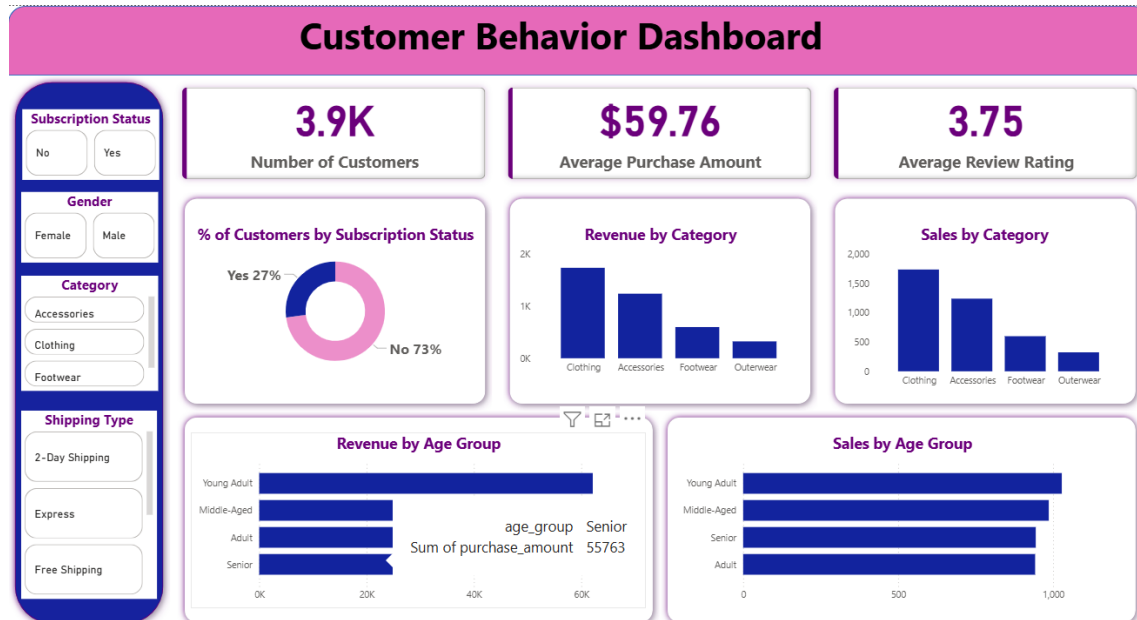
Calculated total revenue contribution of different age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

Young Adult and Middle-Aged groups contribute the highest share of revenue.

5. Dashboard in Power BI

Finally, an interactive dashboard was created using Power BI to present the insights in a visual and easy-to-understand format.



The above dashboard provides a visual summary of customer behaviour, including total customers, average purchase amount, and review ratings. It highlights key insights such as revenue and sales distribution across different product categories and age groups.

The dashboard also includes interactive filters like subscription status, gender, and shipping type, allowing users to explore the data dynamically. This helps in better understanding customer trends and supports data-driven decision-making.

6. Business Recommendation

- **Boost Subscriptions** – Offer attractive benefits and exclusive deals to encourage more customers to subscribe.
- **Customer Loyalty Programs** – Introduce reward programs for repeat buyers to convert them into loyal customers.
- **Optimize Discount Strategy** – Maintain a balance between increasing sales and protecting profit margins.
- **Product Promotion** – Focus on marketing top-rated and high-performing products to increase visibility.
- **Targeted Marketing** – Concentrate on high-revenue customer segments, especially key age groups and premium shipping users.